

Physical Oceanography Lab: M1 Banyuls

Network Setup

- 1) Turn on your computer
- 2) Click “connexion bureau” -> “connexion” -> “connexion” (patience needed !)
- 3) Each student has a username and password which are the same:

e.g.: utilisateur toulouse5
 mot de passe toulouse5

- 4) Find your working directory:

Demarrer -> poste de travail -> disque local (C:) -> dossier commun -> toulouse

at this point you should create a working directory within your group:

e.g.: toulouse/group1/yourname

you will find your data under toulouse/Seabird

you should copy this to your working directory

IMPORTANT

any time you copy or move anything *anywhere* use copy paste, *not* drag and drop.

- 5) start the program you want to use

examples:

Demarrer -> tous les programmes -> sea-bird -> SBEDataProcessingWin32

or

Demarrer -> tous les programmes -> scilab5.1.1 -> scilab5.1.1

Seabird

For analysing and visualising CTD data

Start by copying the following files to your working directory:

raw data (e.g.: 20090610a.hex);
the configuration file (e.g.: 20090610a.CON);
the program setup file for the process you are about to use (e.g.: DatCnv.psa).

These files can all be found in toulouse/Seabird.

Then you can process the data in your working directory.

In the Seabird interface you need to set up pathnames and options.

So to run a process (e.g. "filter"):

- *open the .psa file that you have copied to your workspace: OPEN filter.psa*
- *select your input file: SELECT <last_file_you_worked_on>.cnv*
- *select output directory: SELECT your workspace, append _filter, keep output filename same as input filename.*

In each case make sure the path name to your working file points to your working directory !

When you've set all this up push START PROCESS.

Most of the processes will take a file like myfile.cnv and add your appendix to create for example myfile_filter.cnv. This way you keep a different file for each stage of the process.

The first program to run is datcnv which takes myfile.hex and creates myfile.cnv

At any time you can look at a file using bloc note (right click -> open with -> block note). You can use this to edit files too. Important: *don't* double click on a .cnv file ! Windows will open it with something inappropriate.

At any time you can also use Sea Plot to visualise the results.

Scilab

Some scilab routines are available in the directory SEAWATER to diagnose the properties of your profiles step by step and to calculate the vertical profile of the geostrophic current.

Again, copy them to your working directory before use.

To work on profile data produced with Seabird, you'll need to edit the file to remove the header to create an ascii file with just columns of data.

Once you've launched the scilab interface, here are some simple scilab commands to get started:

Read a profile:

```
a = read('test_profile.txt',-1,3)
```

Partition the variables:

```
p = a(:,1) ; (reads the first column into a variable p)
T = a(:,2) ;
S = a(:,3) ;
```

Plot the variables:

```
plot (T,-p) (plots temperature as a function of depth (pressure) increasing downwards)
```

Calculate new variables:

For this you can use ready-made scilab routines. For example you can calculate the reference density (at surface pressure) from S and T. You'll need to define two functions, one to calculate the standard density of mean ocean water as a function of temperature (sw_smow.sci) and the other to calculate the reference density (sw_dens0.sci). Do this:

```
exec sw_smow.sci
exec sw_dens0.sci
```

then define the new variable

```
D0 = sw_dens0(S,T)
```

Save data (for display with other software):

```
savematfile fn.asc var' - ascii
```

makes an ascii file with a column format you can cut and paste into excel

otherwise (better) you can do

```
print('fn.asc',var1',var2')
```

If you want to print or save graphs from the graphics window do:

fichier -> exporter vers -> (file type: png etc)

Density of seawater: Scilab calculations

Before we go on, here is a rundown of the way we calculate physical variables for sea water:

Start with standard density of mean ocean water

$$\rho_s \quad (S = 0, p = p_0)$$

This is just a function of temperature

$$\rho_s = a_0 + a_1T + a_2T^2 + a_3T^3 + a_4T^4 + a_5T^5$$

then add salinity effects

$$\rho_0 = \rho_s + (b_0 + b_1T + b_2T^2 + b_3T^3 + b_4T^4)S + (c_0 + c_1T + c_2T^2)S^{1.5} + d_0S^2$$

then add compressibility effects (the effect of pressure)

$$\rho = \rho_0(1 - p/\kappa)$$

where kappa is the bulk modulus (the reciprocal of the compressibility)

$$\kappa = \frac{\Delta p}{\Delta V/V}$$

List of scilab routines

all with format sw_name.sci

<i>adtg:</i>	<i>adiabatic temperature gradient (°C/dbar)</i>
<i>bfrq:</i>	<i>Brunt Vaisala frequency squared (N²) and stratification pot. vorticity f N²/g</i>
<i>dens:</i>	<i>density from (S,T,p)</i>
<i>dens0:</i>	<i>density at atmospheric pressure from (S,T)</i>
<i>dist:</i>	<i>distance between two stations given (latitude, longitude)</i>
<i>pden:</i>	<i>potential density with respect to specified reference pressure (S,ptmp,pR)</i>
<i>ptmp:</i>	<i>potential temperature with respect to reference pressure (S,T,p,pR)</i>
<i>seck:</i>	<i>secant bulk modulus (S,T,p)</i>
<i>smow:</i>	<i>standard density of mean ocean water (T)</i>
<i>svan:</i>	<i>specific volume anomaly with respect to (35 psu, 0°C, p): $\Delta(1/\rho)$ (S,T,p)</i>

Calculating the geostrophic velocity (relative to “level of no motion”)

use the scilab routine

geovel: *geostrophic velocity with respect to reference pressure*

You have to set up the input data and initialise some functions, but to make things easy for you we've written a script so all you have to do is prepare the input data, make sure all the necessary code is in your working directory, and type

`exec geovel_shallow_ex.sci`

you'll have to edit this exec first to make sure it reads the right files:

Define your reference pressure (level of no motion) in dbar (eq. metres).

Insert relevant filenames for your two stations (make sure the header has been removed from your ascii files).

Define station 1 and station 2 by their latitude and longitude (lat1,lon1,lat2,lon2).

The calculation carried out by the code is as follows:

Hydrostatic relation

$$\frac{\partial p}{\partial z} = -\rho g, \quad \frac{\partial p}{\rho} = -g \partial z, \quad g z = - \int \frac{dp}{\rho} = - \int SV dp$$

(SV stands for Specific Volume)

Geostrophic relation

$$f v = \frac{1}{\rho} \nabla p = -g \nabla z = -\nabla \int SV dp$$

NOTE: The program basically calculates u_g , the geostrophic zonal velocity, assuming the stations are placed south-north. Specifically station 1 should be to the south and station 2 should be to the north. The geostrophic current flows to the right of the path from station 1 to station 2, i.e. positive if it is eastward and negative if it is westward.

If your stations are oriented west-east, the same calculation is made, which means that you should define station 1 to the east and station 2 to the west. This way the result will be positive for a northward current and negative for a southward current.